

1. We are developing an **online T- shirt sales system**. The total order cost is calculated based on the following two factors:

Product category:

Basic T- shirt
Printed T- shirt
Premium T- shirt

Order quantity:

Small (1–50 T- shirts)
Medium (51–100 T- shirts)
Large (101 or more T- shirts)

Cost calculation rules:

Base price (per T- shirt)

Basic T- shirt: 20 RMB
Printed T- shirt: 45 RMB
Premium T- shirt: 120 RMB

Delivery fee (based on order quantity)

Small or Medium order (≤ 100 T- shirts): 10 RMB
Large order (≥ 101 T- shirts): 0 RMB

Total cost = Quantity $\times$ Base price + Delivery fee

Question 1: Design a decision table for the cost calculation logic of this T- shirt sales system.

Question 2: Based on the decision table and the cost calculation rules, design Normal Boundary Value Testing cases for the system.

Solution

1. Decision Table

Conditions

- C1: Product category = Basic
C2: Product category = Printed
C3: Product category = Premium
C4: Quantity ≤ 100 (Small or Medium)
C5: Quantity ≥ 101 (Large)

Actions

- A1: Unit price = 20 RMB
A2: Unit price = 45 RMB
A3: Unit price = 120 RMB
A4: Delivery fee = 10 RMB
A5: Delivery fee = 0 RMB
A6: Total cost = Quantity $\times$ Unit price + Delivery fee

decision table (Y/N format)

Condition / Rule	1	2	3	4	5	6
C1: Basic T-shirt?	Y	Y	N	N	N	N
C2: Printed T-shirt?	N	N	Y	Y	N	N
C3: Premium T-shirt?	N	N	N	N	Y	Y
C4: Quantity ≤ 100 ?	Y	N	Y	N	Y	N
C5: Quantity ≥ 101 ?	N	Y	N	Y	N	Y
Actions						
A1: Unit price = 20	X	X				
A2: Unit price = 45			X	X		
A3: Unit price = 120					X	X
A4: Delivery fee = 10	X		X		X	
A5: Delivery fee = 0		X		X		X
A6: Compute total cost	X	X	X	X	X	X

decision table (explicit values)

Rule	Product category	Quantity range	Unit price (RMB)	Delivery fee (RMB)	Total cost formula
1	Basic	1-50	20	10	$20 \times Q + 10$
2	Basic	51-100	20	10	$20 \times Q + 10$
3	Basic	≥ 101	20	0	$20 \times Q$
4	Printed	1-50	45	10	$45 \times Q + 10$
5	Printed	51-100	45	10	$45 \times Q + 10$
6	Printed	≥ 101	45	0	$45 \times Q$
7	Premium	1-50	120	10	$120 \times Q + 10$
8	Premium	51-100	120	10	$120 \times Q + 10$
9	Premium	≥ 101	120	0	$120 \times Q$

6.2 Normal Boundary Value Testing Cases

Boundary identification:

Small quantity boundaries: 1 (min), 50 (max)

Medium quantity boundaries: 51 (min), 100 (max)

Large quantity boundaries: 101 (min, also the delivery - fee cutoff) and a typical large value (e.g., 150)

Critical boundary between “with delivery fee” and “no delivery fee”: 100 vs 101

Normal Boundary Value Testing uses only valid boundary values from each equivalence class (no invalid inputs like 0 or negative numbers).

Test cases

Test ID	Product category	Quantity	Expected unit price (RMB)	Expected delivery fee (RMB)	Expected total cost (RMB)	Boundary covered
TC1	Basic	1	20	10	$20 \times 1 + 10 = 30$	Small – lower bound
TC2	Basic	50	20	10	$20 \times 50 + 10 = 1010$	Small – upper bound
TC3	Basic	51	20	10	$20 \times 51 + 10 = 1030$	Medium – lower bound
TC4	Basic	100	20	10	$20 \times 100 + 10 = 2010$	Medium – upper bound (≤ 100)
TC5	Basic	101	20	0	$20 \times 101 = 2020$	Large – lower bound (≥ 101)
TC6	Basic	150	20	0	$20 \times 150 = 3000$	Large – typical value
TC7	Printed	1	45	10	$45 \times 1 + 10 = 55$	Small – lower bound
TC8	Printed	50	45	10	$45 \times 50 + 10 = 2260$	Small – upper bound
TC9	Printed	51	45	10	$45 \times 51 + 10 = 2305$	Medium – lower bound
TC10	Printed	100	45	10	$45 \times 100 + 10 = 4510$	Medium – upper bound
TC11	Printed	101	45	0	$45 \times 101 = 4545$	Large – lower bound
TC12	Printed	150	45	0	$45 \times 150 = 6750$	Large – typical value
TC13	Premium	1	120	10	$120 \times 1 + 10 = 130$	Small – lower bound
TC14	Premium	50	120	10	$120 \times 50 + 10 = 6010$	Small – upper bound
TC15	Premium	51	120	10	$120 \times 51 + 10 = 6130$	Medium – lower bound
TC16	Premium	100	120	10	$120 \times 100 + 10 = 12010$	Medium – upper bound
TC17	Premium	101	120	0	$120 \times 101 = 12120$	Large – lower bound
TC18	Premium	150	120	0	$120 \times 150 = 18000$	Large – typical value

Explanation:

Each product category (Basic, Printed, Premium) is tested with all key boundary values: 1, 50, 51, 100, 101, and a representative large quantity (150).

The critical boundary (100 vs 101) verifies that the delivery fee changes from 10 RMB to 0 RMB correctly.

All inputs are valid (quantity ≥ 1), making these Normal boundary value test cases.

The test cases fully cover all six rules in the decision table.

Consider the following C function adjustValue(int x, int y, int mode).

```
int adjustValue(int x, int y, int mode) {  
    int result = 0;  
  
    // First if-else statement  
    if (x > y) {                                // (1)  
        result = result + 10;                    // (2)  
    } else {  
        result = result + 5;                    // (3)  
    }  
  
    // Second if-else statement (separate)  
    if (mode == 1) {                             // (4)  
        result = result * 2;                    // (5)  
    } else if (mode == 2) {                      // (6)  
        result = result + 20;                  // (7)  
    } else {  
        result = result - 5;                   // (8)  
    }  
  
    return result;                             // (9)  
}
```

Question 1: Draw a control flow graph corresponding to the function above.

Question 2: Derive a minimal set of test cases that can achieve branch coverage (also known as decision coverage) based on the control flow graph above.

Question 3: Derive a minimal set of test cases that can achieve McCabe's Basis Path Testing coverage based on the control flow graph above

Number of nodes $N=9$ (labeled 1 to 9)

Number of edges $E=11$

Cyclomatic complexity $V(G)=E-N+2=11-9+2=4$

A set of linearly independent paths (basis set) covering all edges/nodes:

Path ID	Description	Path (node sequence)
P1	$x > y, \text{ mode} == 1$	$1 \rightarrow 2 \rightarrow 4 \rightarrow 5 \rightarrow 9$
P2	$x > y, \text{ mode} == 2$	$1 \rightarrow 2 \rightarrow 4 \rightarrow 6 \rightarrow 7 \rightarrow 9$
P3	$x > y, \text{ mode} == \text{other } (\neq 1, \neq 2)$	$1 \rightarrow 2 \rightarrow 4 \rightarrow 6 \rightarrow 8 \rightarrow 9$
P4	$x \leq y, \text{ mode} == 1$	$1 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 9$